Geometric Distribution

The Geometric Distribution is the distribution of the number of trial needed to get the first success in repeated bernouli trials.

Assumption

- There are independent trial
- each trial result in one of two possible outcome, labelled success and failure

P(Success) = p, and this stays constant from trial to trial
P(Failure) = 1 - p
X = the number of trials needed to get the first success

Probability Mass Function

$$P(X = x) = (1 - p)^{x-1}p$$

for x= 1,2,3.....

- The first trials must be failure
- the xth trial must be a success

$$\mu = \frac{1}{p}$$

$$\sigma^2 = \frac{1-p}{p^2}$$

1st Example:

In a large population of adults, 30% have received CPR training.

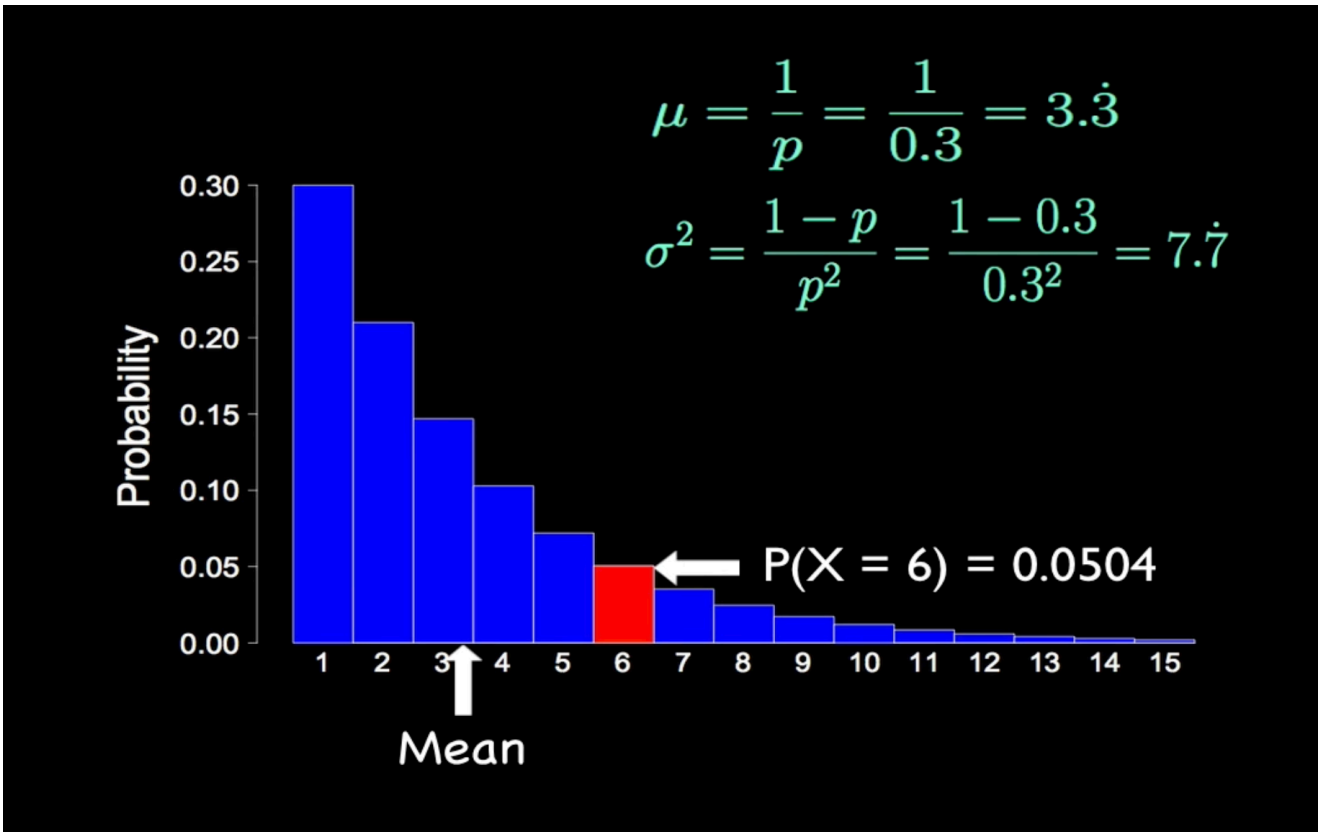
$p = 0.3$

If adults from this population are randomly selected, what is the probability that the 6th person sampled is the first that has received CPR training?

$$P(X = x) = (1-p)^{x-1}p$$

$$P(X=6) = (1-0.3)^5 0.3$$

$$= 0.0504$$

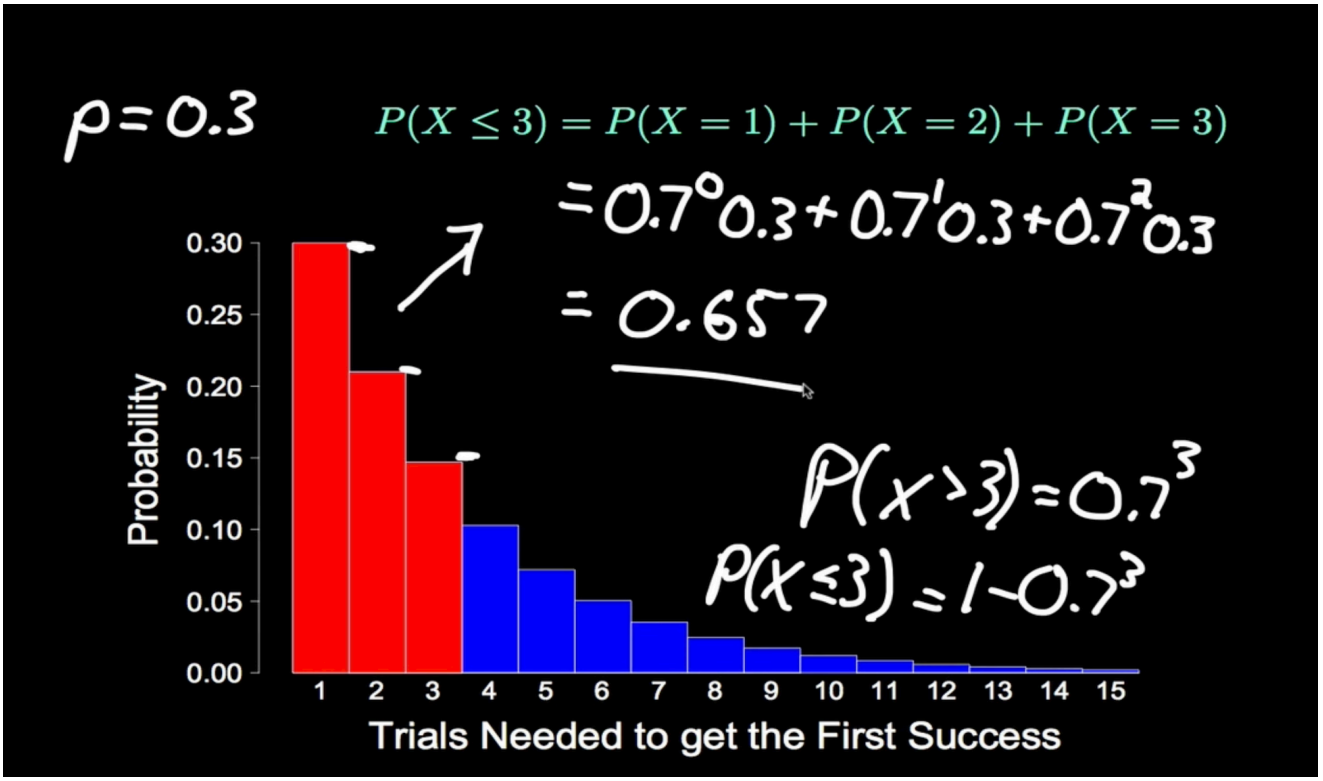


What is the probability that the first person trained in CPR occurs on or before the 3rd person sampled?

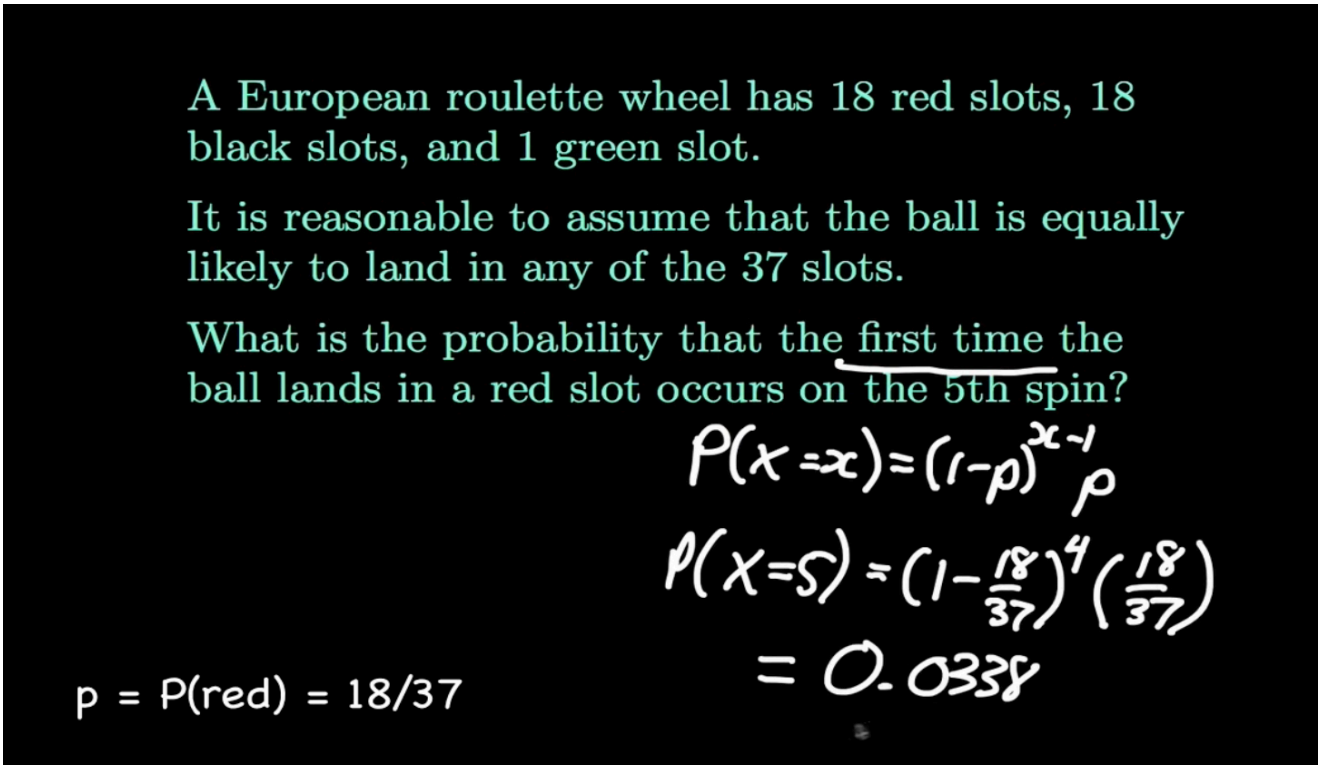
p(x=<3)

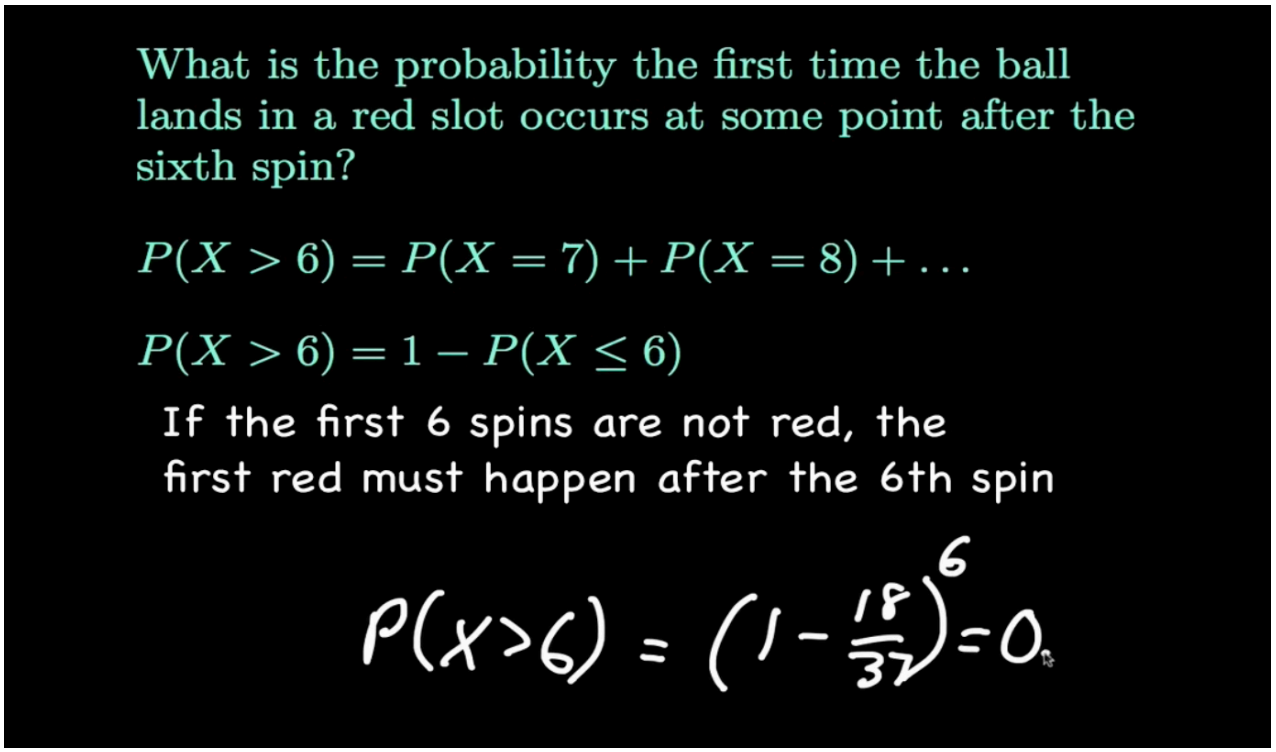
--> **Cumulative distribution function for the geometric distribution**

$$F(x) = P(X \leq x) = 1 - (1-p)^x$$


p=0.3
P(X ≤ 3) = P(X = 1) + P(X = 2) + P(X = 3)
=0.7^0 0.3+0.7^1 0.3+0.7^2 0.3
= 0.657
P(X>3)=0.7^3
P(X≤3) = 1-0.7^3
Probability
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Trials Needed to get the First Success

2nd Example:


A European roulette wheel has 18 red slots, 18 black slots, and 1 green slot.
It is reasonable to assume that the ball is equally likely to land in any of the 37 slots.
What is the probability that the first time the ball lands in a red slot occurs on the 5th spin?
P(X=x)=(1-p)^{x-1} p
P(X=5)=(1-18/37)^4 (18/37)
= 0.0338
p = P(red) = 18/37



Credit to the owner